# Ping Wang

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## Research Interests

I want to build AI systems that stay aligned with people and keep improving as they interact with them. I study reinforcement learning and post-training for generative models, and the preference learning, reward modeling, and adaptive evaluation that make that training trustworthy. I am also interested in how these ideas extend to multi-agent and social learning, human–AI coordination and co-evolution, and adaptive learning environments and world models. Music, a multimodal domain where quality is hard to specify, is my main testbed.

## Education

**University of Washington** Seattle, WA
*B.S., Paul G. Allen School of Computer Science & Engineering; GPA: 3.91/4.0* *Jun 2027*

- **Relevant Coursework**: Machine Learning, Natural Language Processing, Social Reinforcement Learning, Probabilistic Graphical Model, Computer Graphics, Data Structures and Algorithms, Distributed Systems, Operating Systems, and Computer Networks.

## Publications and Preprints

**Ping Wang**, Fang-Duo Tsai, Wei-Jaw Lee, and Yi-Hsuan Yang. *SheetSage-Pro: Bilingual Lyric-Aligned Lead-Sheet Transcription from Music Audio.* ISMIR 2026 Late-Breaking/Demo. [Paper] [Code].

**Ping Wang**, Guang Yang, Shao-Rong Su, Junkai Wu, Pang Wei Koh, and Noah A. Smith. *Rubric-Based Optimization for Text-to-Music Generation.* Under review at ICLR 2027. [arXiv].

**Ping Wang**, Guang Yang, Nazif Can Tamer, Victoria Ebert, and Noah A. Smith. *A Dual Evaluation for Music Transcription.* Under review at TISMIR, 2026. [arXiv].

## Research Experience

**Noah's Ark Lab, University of Washington** Seattle, WA
*Student Researcher — Advisor: Prof. Noah A. Smith* *Aug 2025 – Present*

- **A Dual Evaluation for Music Transcription**: Developed a dual evaluation framework for assessing both notation accuracy and performance fidelity in automatic music transcription. Evaluated 24 transcription pipelines composed of eight audio-to-MIDI models and three MIDI-to-score systems across 230 piano performances. Designed and analyzed an IRB-exempt, audio-only listening study: a reference-anchored two-alternative forced-choice (ABX) task over all 276 system pairs, with A/B order randomized and no scores shown, 30 trials per participant, and recruitment sized by a simulation-based power analysis (106 participants, 3,180 judgments). Aggregated judgments with a Bradley–Terry model and used them to validate DTW, TWED, CLaMP 3, CLEWS, and Gemini judges (run in both A/B orders to cancel position bias); CLEWS agreed best with listeners (Spearman $\rho = 0.971$) at the lowest cost.
- **Rubric-Based Optimization for Text-to-Music Generation**: Developed structured rubric rewards from pretrained audio-language models for text-to-music post-training. Scored candidates generated for the same prompt against the rubric and turned the rankings into preference pairs for direct preference optimization (DPO) on MusicGen-small and ACE-Step v1, and used the rubric scores directly as scalar rewards for online RL with DiffusionNFT on ACE-Step. On MusicCaps, rubric rewards improved CLAP, SongEval, and Audiobox-Aesthetics together, while any single automatic metric used as the reward caused cross-metric trade-offs; specialized objective rewards remained better for precise attributes such as tempo, key, and instrumentation.

**Music and AI Lab, National Taiwan University** Taipei, Taiwan
*Student Researcher — Advisor: Prof. Yi-Hsuan Yang* *May 2026 – Present*

- **SheetSage-Pro**: Built a bilingual (Mandarin/English), lyric-aligned audio-to-lead-sheet transcription system that integrates pretrained models for source separation, beat and downbeat tracking, structure analysis, melody, chords, and lyric alignment. Against the original Sheet Sage, scored with the same code, it raises note F1@50 ms on GTSinger from 0.080 to 0.295 and chord WCSR on GuitarSet from 0.621 to 0.904, and its melody component reaches 0.670 on MIR-ST500. **Accepted for presentation at ISMIR 2026 Late-Breaking/Demo.**

- **Lyrics-to-Song Generation with MIDI-LLM and MIDI-SAG**: Developing an end-to-end lyrics-conditioned song generation pipeline by connecting symbolic lead-sheet generation with downstream arrangement and audio synthesis. Fine-tuning language models to generate melody, chords, rhythm, and lyric-aligned symbolic representations, including syllable-level lyric pointers, phonetic information, rests, breaths, and many-to-many lyric–note alignment for controllable generation.

**UbiComp Lab, University of Washington** Seattle, WA
*Student Researcher — Advisor: Prof. Shwetak Patel* *Sep 2024 – Oct 2025*

- **Embedded Gesture Recognition**: Designed and optimized lightweight neural networks for real-time gesture recognition on resource-constrained embedded devices, including model quantization and deployment for on-device inference.
- **Tuberculosis Screening from Cough Audio**: Applied transformer-based architectures, including Audio Spectrogram Transformer (AST) and MobileViT, to cough-based tuberculosis risk classification.
- **Cough Detection and Localization**: Developed machine learning models for cough-event detection and temporal localization in continuous audio recordings.

## Industry Experience

**Amazon** Taipei, Taiwan
*Software Development Engineer Intern* *Jun 2025 – Sep 2025*

- **Embedded Firmware Development**: Designed and implemented embedded firmware for inter-chip communication, camera-module integration, and Wi-Fi connectivity validation for smart-home devices.
- **Hardware–Software Integration**: Worked on board-level integration and debugging across embedded hardware, firmware, camera modules, and network interfaces.
- **Testing Infrastructure**: Developed automated unit tests and peripheral mock frameworks for validating firmware across hardware states, improving test coverage and reducing regression risk.

**Phison Electronics** Miaoli, Taiwan
*Machine Learning / Embedded Systems Engineer Intern* *Jun 2024 – Aug 2024*

- **LLM Memory Modeling**: Developed a parameter-calculation system for estimating large language model memory consumption, including model weights and activations, directly from model configuration files.
- **Transformer Architecture Analysis**: Analyzed LLaMA 2/3, CodeLlama, Falcon, Mistral, Mixtral, Breeze, Whisper, and Vicuna, with emphasis on attention architectures, model scaling, and memory requirements.
- **Large-Scale Model Training**: Worked with multi-GPU Linux infrastructure and Phison AI SSD systems for model training and experimentation.

## Technical Skills

**Programming Languages**: Python, C, C++, Java
**Machine Learning**: PyTorch, Transformers, Preference Optimization (DPO), Reinforcement Learning, Diffusion / Flow-Matching Models, LoRA
**Music and Audio AI**: Automatic Music Transcription, Music Generation, Audio Representation Learning, Symbolic Music Modeling, MIDI, MusicXML
**Systems and Tools**: Linux, Git, Distributed Training, Multi-GPU Computing, Embedded Systems, FPGA